image. The tenth image is very near the left edge of the shadow and the right secondary image is near the right edge; comparing the directions of the camera for these images, we see that the shadow subtends about 150 degrees in the upward direction. This despite the fact that the actual direction from camera to center of Gargantua is leftward and upward. The lensing has moved the shadow relative to Gargantua's actual direction.

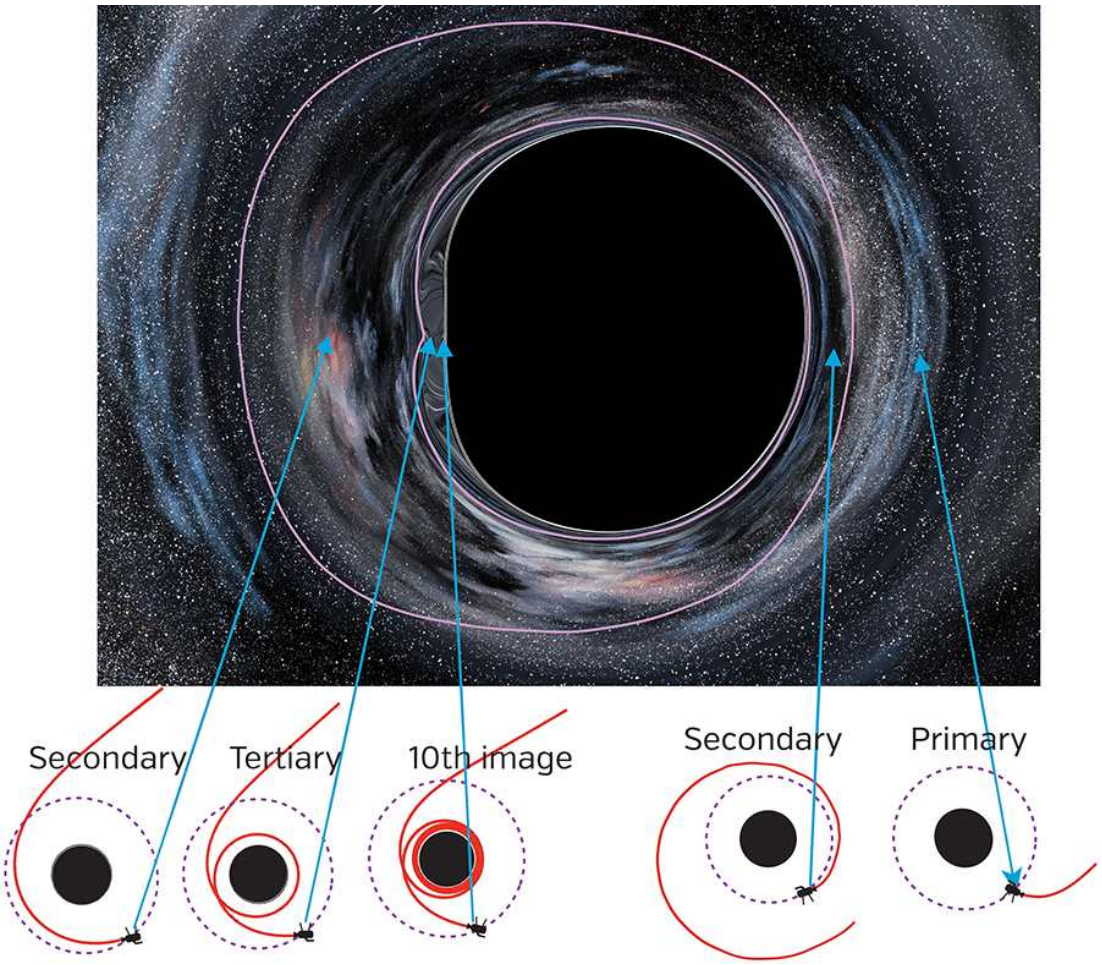


Fig. 8.6. Light rays that bring images of the stars at the tips of the blue arrows.
[From the same Double Negative simulation as Figures 8.1 and 8.5.]

Creating *Interstellar*'s Black-Hole and Wormhole Visual Effects